

# **Problem Statement : Aspects of Vaccines**

**Course Name : Biology**

**Course Code : BSC 401**

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**Semester: IV**

**Program Name: B. Tech. (IT)**



**Continuous Assessment - 2 (CA2)**

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### • Definition:

Vaccines are biological preparations that provide active acquired immunity to a particular infectious disease. They act as "training sessions" for the immune system, allowing the body to recognize and fight pathogens without the individual having to experience the full-blown illness.

### Types of Vaccines

- Live Attenuated Vaccines - Contain weakened forms of live virus or bacteria

Example: Measles, Mumps.

- Inactivated Vaccines: - Contain dead or killed pathogens

Example: polio vaccine

- Subunit Vaccines - Contain specific pieces of pathogen (proteins or antigens)

Example: Hepatitis B

- M-RNA Vaccines: Contain genetic instructions to produce viral proteins  
Example: Covid-19 Vaccine

- How Vaccines Work

- Vaccine enters the body and presents antigen to immune system.
- B-cells recognize antigen and produce antibodies
- T-cells are activated to coordinate immune response.
- If exposed to real pathogen, immune system quickly recognizes and eliminates
- Process typically takes 1-2 weeks for initial response.

## • Development

Pre-Clinic Stage: Testing in cell cultures on animal model to access basic Safety.

Phase-I: Small group to evaluate

Phase-II: Expanded groups to study efficacy and Common Side Effect.

Phase-III: Large Scale to Confirm effectiveness and monitor for rare adverse.

## Vaccination Schedules

- Childhood Vaccines: Given from birth to age 18
  - Include: Polio, Measles, Diphtheria etc

- Adolescent Vaccines: For age 11-18 years

- Adult Vaccines: For individuals 18+ years

- Booster Shots - Additional doses to maintain Immunity after initial Vaccination.



## Benefits Of Vaccination

- Prevent Serious Diseases - Elimination or reduces risk of infectious diseases
- Herd Immunity - Protects Vulnerable populations (Infants, elderly)
- Cost Effective - Prevention is cheaper than treating disease complications
- Reduces Healthcare Burden - Fewer hospitalizations
- Global Disease Eradication
- Saves Lives.

## Some Side Effects

- Common Side Effects
  - pain, redness
  - fever
  - Muscle aches
- Rare Serious Side Effect
  - Severe allergic reaction
  - Guillain-Barré syndrome
  - Myocarditis.

## Resources And References

- World Health Organization - Vaccines Section  
• Website : [www.who.int/health-topic](http://www.who.int/health-topic)
- Centers for Disease Control and Prevention (CDC)  
Website : [www.cdc.gov/vaccines](http://www.cdc.gov/vaccines)
- National Institute of Health (NIH)
- Gavi - The Vaccine Alliance.